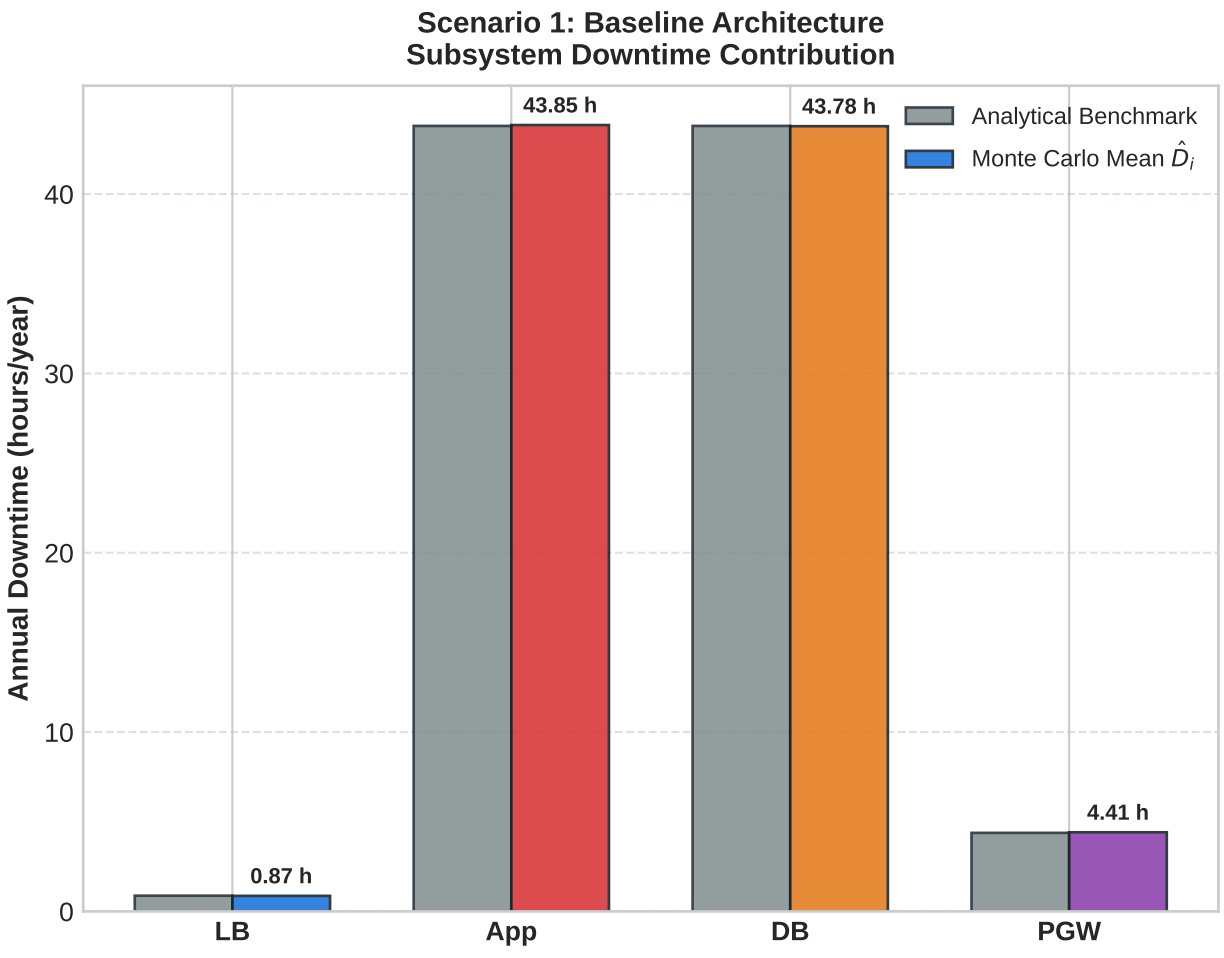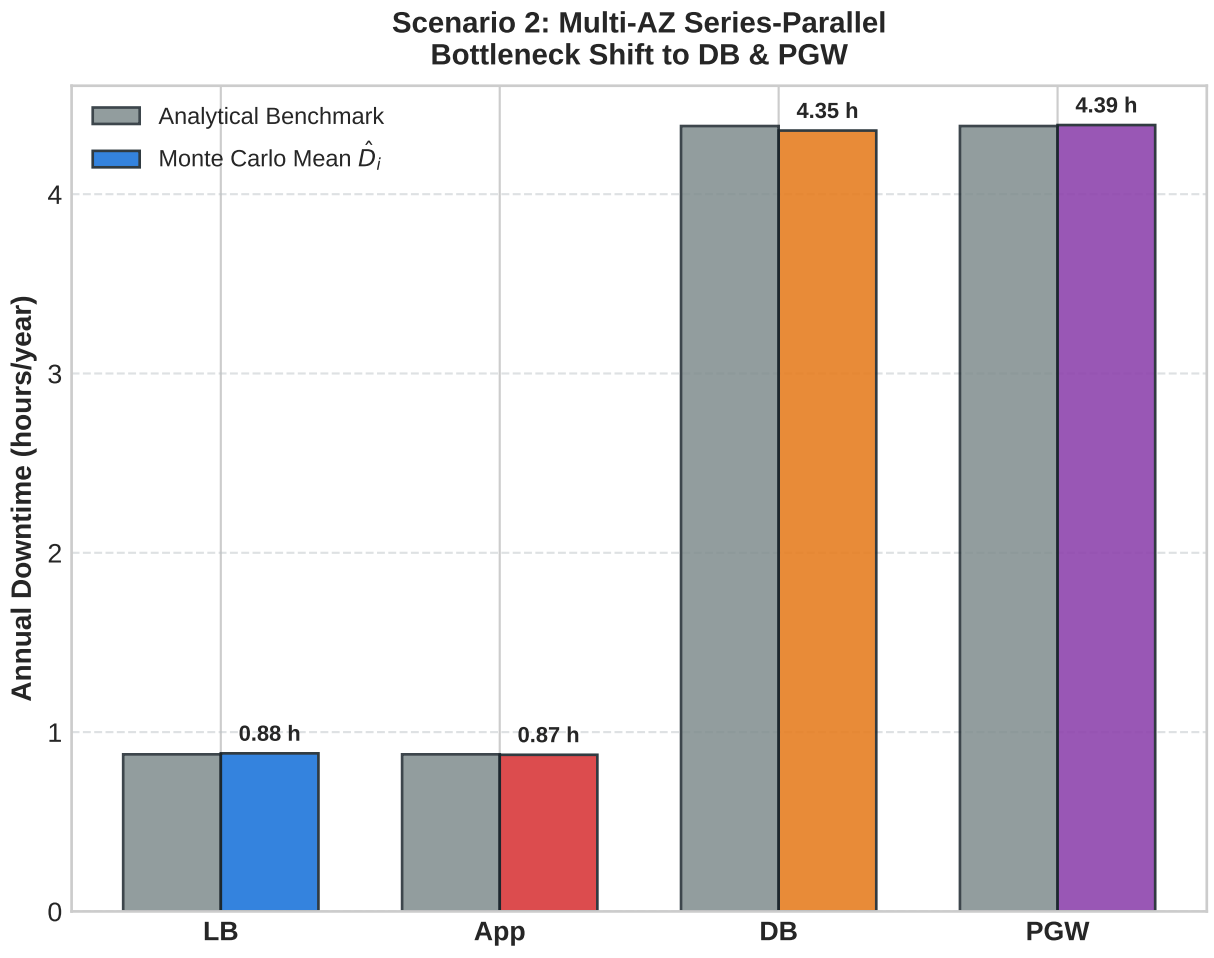Monte Carlo Validation of Subsystem Downtimes and Architectural Bottleneck Shift